

31-Rossman Sales Prediction Project

Introduction

- Context: Classroom lecture focused on a practical machine learning project—Rossman sales prediction.
- Objective: Predict sales for Rossman stores using store promotion and competitor data, highlighting EDA and basic data preparation.

Dataset Overview

- Four uploaded files: three CSV datasets (store, train, test) and one code (ipynb) file.
- Store data: Contains variables such as store type, assortment, competition distance, competition open since month/year, promo2, promo2 since week/year, and promo interval.
- Train data: Includes store identifier, day of week, date, sales, customers, open, promo, state holiday, school holiday.
- Test data: Contains store identifiers and relevant activity metrics.

Data Loading & Merging

- Data files are loaded using pandas' read_csv method.
- Store data and train data are merged on the store ID to enrich the training set with store-specific features.
- The merging process aligns information for each store across different dates and sales activities.

- Verification of data shapes, unique values, and intersection of columns performed.

Exploratory Data Analysis (EDA)

- Identification and handling of null values across datasets.
- Review of data types in each column (ints, floats, objects/strings).
- Conversion recommendations: date to datetime, holiday and promo flags to categorical/int types.
- Extraction of year and month from the date for periodic analysis.
- Identification and listing of unique values in key columns to distinguish categorical features from continuous ones (e.g., sales, customers).

Data Visualization

- Various bar plots and analyses performed to reveal relationships:
 - **Sales vs Year:** Sales patterns over multiple years.
 - **Sales vs Weekday:** Patterns show higher sales on weekdays, lowest on Sundays.
 - **Sales vs Promo Participation:** Promotions positively affect sales.
 - **Sales vs State Holiday**
: Lower sales during state holidays, with non-holiday days yielding higher sales.
 - **Sales vs School Holiday:** Higher sales during school holidays.

- **Sales vs Store Type:** Store type B shows highest sales compared to other types (A, C, D).
- **Sales vs Assortment:** Assortment type B achieves more sales; type A and C less so.
- **Sales vs Competition Distance**
: Locations with more nearby competitors often have higher sales, possibly due to location density and customer traffic.

Handling Missing Values

- Strategy varies by column:
 - **Promo2-related columns:** Nulls filled with zero.
 - **Competition Distance:** Nulls replaced with mean competition distance.
 - **Competition Open Since Month/Year:** Nulls filled using modal (most frequent) values.
- Emphasis on not dropping rows to avoid significant data loss, preferring imputation based on column context and distribution.
- All missing values addressed before finalizing the merged dataset for modeling.

Conclusions & Insights

- Weekdays, store promotions, school holidays, store type (B), assortment type (B), and lack of state holidays are all positively correlated with higher sales.
- Sales insights help design promotions and staffing in retail scenarios.
- Proper data cleaning and feature engineering are prerequisites for modeling.

- Outlier treatment, advanced visualization, and statistical testing are noted for future (statistics) classes.

Q&A and Additional Resources

- Questions covered file access, data type conversions, code errors, and project domain (retail).
- Project falls under retail/e-commerce analytics.
- Suggestions for further practice: W3Resource, Kaggle, and recommended books on pandas and numpy, with links or PDF references provided.
- Lecture materials uploaded to LMS; future classes scheduled to continue EDA and modeling.

Ending Notes

- Lecture emphasizes hands-on practice and iterative learning for data analytics projects.
- Next class to cover advanced topics: feature engineering, outlier removal, and deeper statistical analysis.